Power input, protection, bulk capacitance

File: power_input.kicad_sch

Auxiliary supplies: 60V->12V->5V->3.3V

File: regulators.kicad_sch

Three-phase bridge, shunts, motor pads

File: power_stage.kicad_sch

DRV8353S gate driver and current-sen

File: gate_driver.kicad_sch

STM32G474 MCU, clock, debug, status LEDs

File: mcu.kicad_sch

CAN-FD, onboard encoder, AUX encoder ports, motor thermistor

File: comms_sensors.kicad_sch

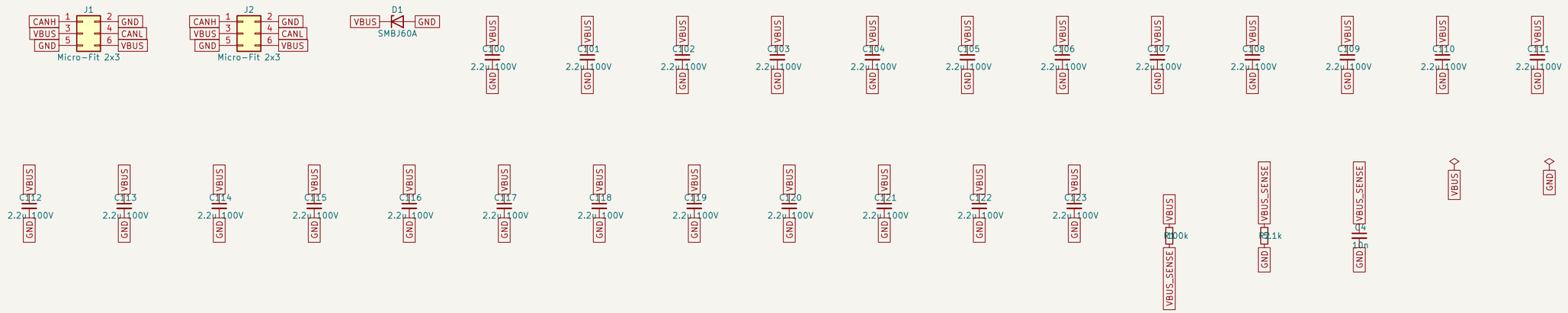
better-md80: MD80 v3.0 form-factor BLDC controller for 48 V nominal / 60 V max buses.

Voltage-limiting parts of the original (gate driver, bus caps, regulators, TVS) replaced with 100 V-class parts.

Connectivity uses global labels: every net name is identical on every sheet.

See ../docs/ for architecture, BOM, thermal/layout and firmware notes.

better-md80		
Sheet: /		
File: better-md80.kicad_sch		
Title: better-md80 - 60 V MD80-compatible motor controller		
Size: A4	Date: 2026-09-28	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/7

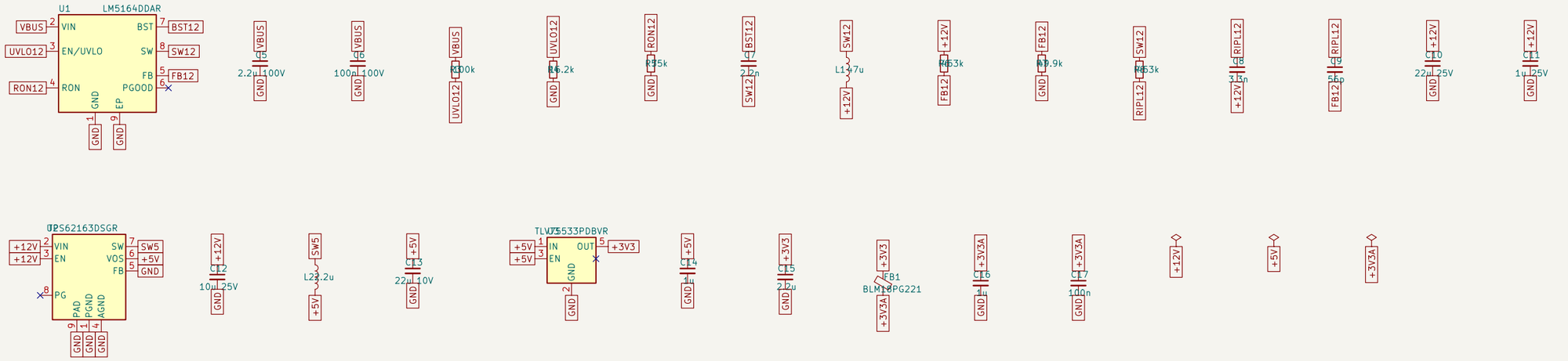


Generated by tools/gen_sch.py from tools/design.py — edit design.py, not this file
48 V nominal / 60 V max MD80-compatible BLDC controller
better-md80

Sheet: /Power input, protection, bulk capacitance/
File: power_input.kicad_sch

Title: Power input, protection, bulk capacitance

Size: A3	Date: 2026-09-28	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/7

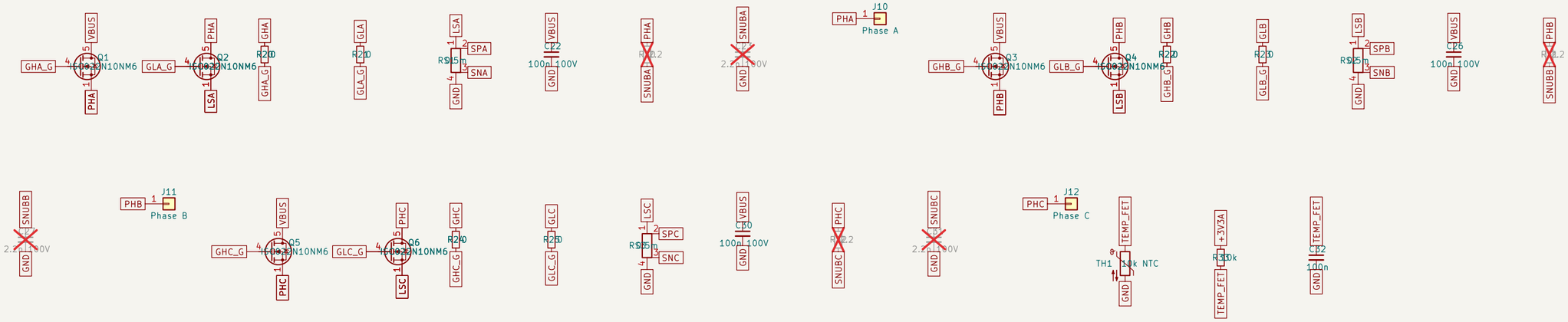


Generated by tools/gen_sch.py from tools/design.py — edit design.py, not this file
48 V nominal / 60 V max MD80-compatible BLDC controller
better-md80

Sheet: /Auxiliary supplies: 60V->12V->5V->3.3V/
File: regulators.kicad_sch

Title: Auxiliary supplies: 60V->12V->5V->3.3V

Size: A3	Date: 2026-09-28	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/7

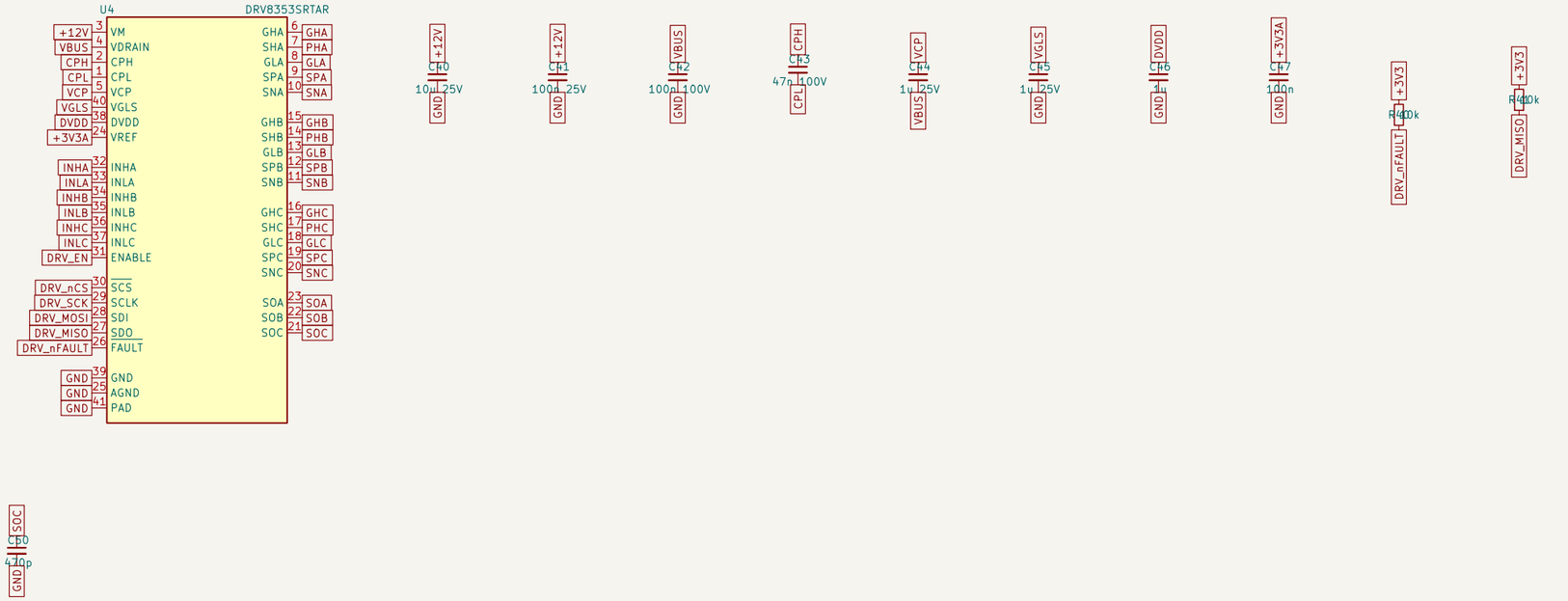


Generated by tools/gen_sch.py from tools/design.py — edit design.py, not this file
48 V nominal / 60 V max MD80-compatible BLDC controller
better-md80

Sheet: /Three-phase bridge, shunts, motor pads/
File: power_stage.kicad_sch

Title: Three-phase bridge, shunts, motor pads

Size: A3	Date: 2026-09-28	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/7



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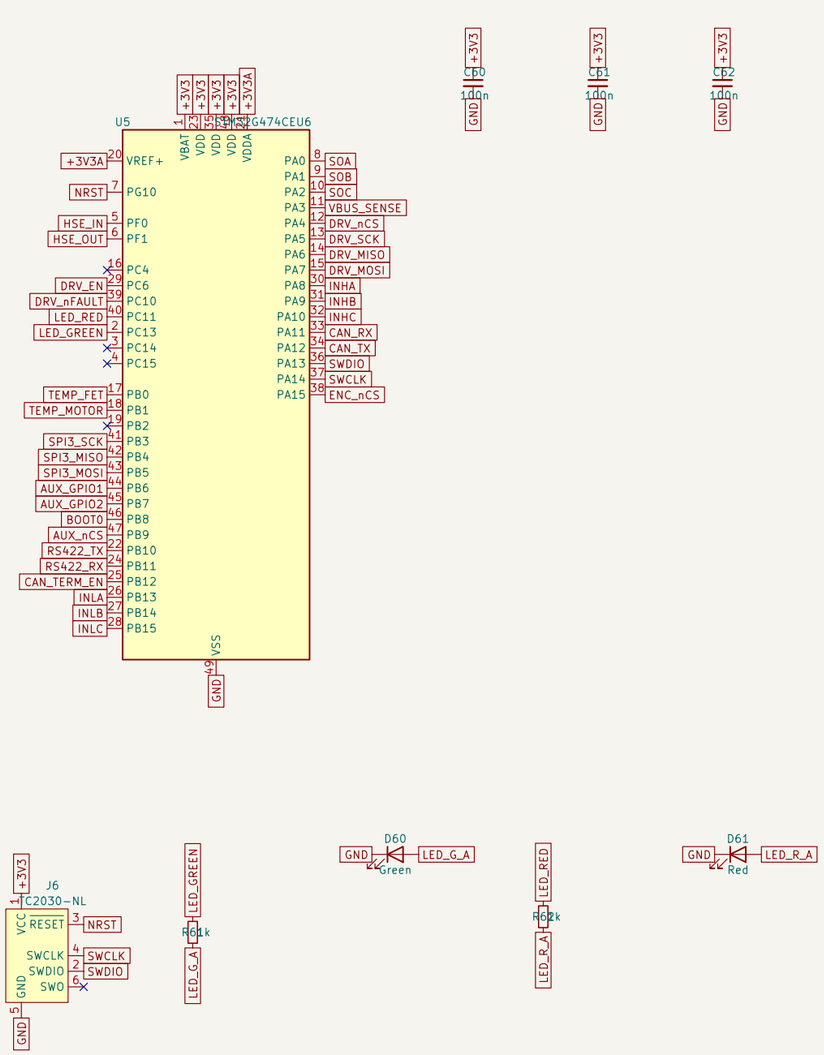
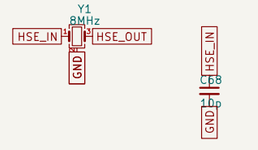
48 V nominal / 60 V max MD80-compatible BLDC controller
better-md80

Sheet: /DRV8353S gate driver and current-sense amplifiers/
File: gate_driver.kicad_sch

Title: DRV8353S gate driver and current-sense amplifiers

Size: A3	Date: 2026-09-28	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/7





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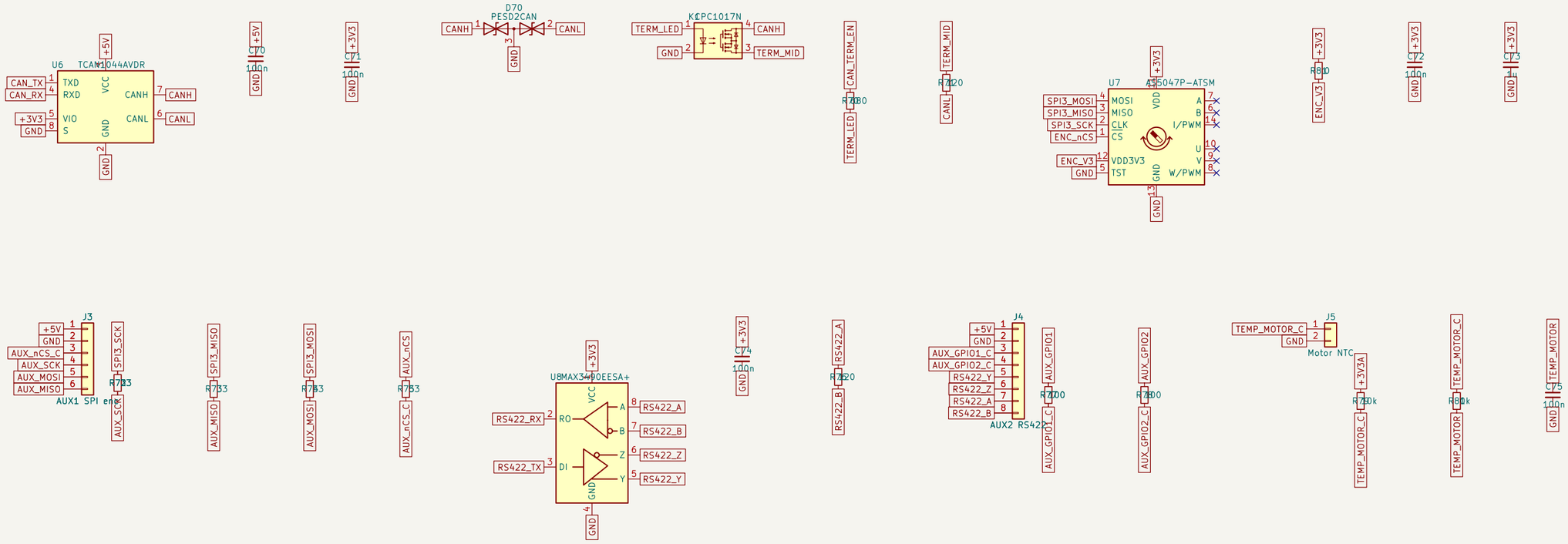
48 V nominal / 60 V max MD80-compatible BLDC controller
better-md80

Sheet: /STM32G474 MCU, clock, debug, status LEDs/
File: mcu.kicad_sch

Title: STM32G474 MCU, clock, debug, status LEDs

Size: A3	Date: 2026-09-28	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/7



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48 V nominal / 60 V max MD80-compatible BLDC controller
better-md80

Sheet: /CAN-FD, onboard encoder, AUX encoder ports, motor thermistor/
File: comms_sensors.kicad_sch

Title: CAN-FD, onboard encoder, AUX encoder ports, motor thermistor

Size: A3	Date: 2026-09-28	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/7